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Lacticaseibacillus rhamnosus OF44 alleviates allergic rhinitis by rebalancing host immunity and gut microbial function

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Allergic rhinitis (AR) involves a maladaptive type 2 inflammatory response driven by systemic immune imbalance and gut dysbiosis. Here, we identify a probiotic strain, *Lacticaseibacillus rhamnosus* OF44, with significant probiotic potential that alleviates allergic pathology and is associated with coordinated immunological and microbial reprogramming. In an ovalbumin-induced AR rat model, OF44 administration markedly reduced nasal allergic symptoms, normalized serum and nasal immunoglobulin and cytokine levels, and restored the balance of Th1/Th2/Th17/Treg cell populations. Metagenomic profiling revealed that OF44 reshaped the gut microbial structure by enriching beneficial commensals (*Rikenellaceae*, *Alistipes*) and suppressing the proinflammatory family Enterobacteriaceae. Functional profiling further demonstrated that OF44 reversed the AR-associated enrichment of pro-inflammatory pathways, including biofilm formation, flagellar assembly, and multidrug resistance, while restoring metabolic pathways related to amino acid metabolism, energy metabolism, and short-chain fatty acid production. Integrated taxonomic–functional correlation analysis suggested that butanoate and lipoic acid metabolic pathways were microbial functions potentially associated with enhanced immune regulation. Collectively, these findings demonstrate that OF44 attenuates AR by reprogramming gut microbial composition and functional capacity, providing mechanistic support for its application as a functional probiotic for the management of allergic disease.

Allergic rhinitis (AR) is an immunoglobulin E (IgE)-mediated inflammatory disorder of the nasal mucosa triggered by inhaled allergens, with typical symptoms including nasal congestion, rhinorrhea, postnasal drainage, sneezing, and itching of the eyes, nose, and throat[1]. These characteristic symptoms significantly impact patients' quality of life, sleep, emotional well-being, and learning performance. It is estimated that AR affects ~10–30% of adults and up to 40% of children, with prevalence rates continuing to increase worldwide, thereby imposing a substantial health and economic burden globally[2,3].

The immunopathology of AR is dominated by type 2 immune responses. During the sensitization phase (the first exposure to allergens without symptoms), dendritic cells in the nasal mucosa capture and process allergens and present them to naive CD4 T cells. Subsequently, naive CD4 T cells become activated and differentiate into allergen-specific T helper type 2 (Th2) cells that induce B-cell differentiation, the production of allergen-specific IgE, and the formation of memory allergen-specific Th2 cells and B cells[4–6]. Upon re-exposure to the relevant allergen, the allergen binds to allergen-specific IgE on mast cells in the nasal mucosa[7]. This triggers mast cell activation and degranulation, releasing both preformed and newly synthesized mediators, including histamine, cysteinyl leukotrienes, prostaglandin D2, and other bioactive mediators. These mediators interact with nasal sensory nerves, blood vessels, and mucosal glands, resulting in the acute symptoms of AR[8].

Current symptomatic treatments for AR, such as antihistamines, intranasal corticosteroids, and allergen immunotherapy, provide clinical benefit for many patients. However, they remain limited by incomplete symptom control in some individuals, potential side effects, limited capacity to modify the underlying disease process, and variable accessibility and

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patient adherence[9]. These limitations underscore the need for adjunctive or alternative strategies that target the underlying immune dysregulation rather than solely alleviating symptoms.

Accumulating evidence indicates that the host microbiota and selected probiotic strains can influence mucosal immunity and the development or severity of allergic airway diseases[10]. Meta-analyses and randomized trials suggest that certain probiotic formulations may improve AR symptoms and patient quality of life; however, substantial heterogeneity among studies (including strain selection, dosage, administration route, and study population) has resulted in inconsistent findings, highlighting the importance of strain-specific evaluation[11,12].

Mechanistically, probiotic bacteria can act on multiple immune axes relevant to allergic responses. They modulate dendritic cell maturation and antigen presentation, shift T-cell differentiation away from Th2 polarization towards regulatory or Th1 phenotypes, enhance regulatory T-cell (Treg) responses, and influence mucosal barrier function and the local cytokine milieus[13]. Beyond immunomodulation, probiotics also exert protective effects by reshaping the gut microbial ecosystem and its metabolic output. By restoring microbial diversity and increasing the abundance of beneficial taxa such as *Lactobacillus* and *Bifidobacterium*, probiotics help re-establish intestinal homeostasis disrupted in allergic conditions[14]. These compositional shifts are often accompanied by altered production of microbial metabolites, including short-chain fatty acids (SCFAs) such as butyrate, acetate, and propionate. These metabolites can regulate epithelial barrier integrity, suppress pro-inflammatory signaling pathways, and promote Treg differentiation through epigenetic regulation and G-protein–coupled receptor signaling[15,16]. Additionally, probiotics influence bile acid and tryptophan metabolism, generating bioactive molecules that signal through host receptors to attenuate mucosal inflammation and reinforce systemic immune tolerance[17]. These host–microbe interactions provide a mechanistic basis for probiotic-mediated attenuation of type 2 airway inflammation.

Lacticaseibacillus rhamnosus strains are among the most studied probiotics in allergic disease models. Preclinical studies have demonstrated that *L. rhamnosus* strains can reduce airway eosinophilia, allergen-specific IgE production, and type 2 cytokine levels in murine models of allergic airway inflammation[18]. In addition, several clinical studies have reported improvements in symptoms of AR or rhinoconjunctivitis following treatment with selected *L. rhamnosus* strains or multi-strain probiotic formulations[19,20]. Moreover, recent work suggests that local (nasal) colonization or topical administration of *L. rhamnosus* may enhance local immune modulation in the upper airway[21].

Despite these promising data, probiotic effects are highly strain-specific and depend on administration route, dosage, and host context; therefore, each newly isolated candidate strain requires rigorous in vivo evaluation to establish its efficacy and mechanism of action. *L. rhamnosus* OF44 has been shown to possess significant probiotic potential. However, previous studies on *L. rhamnosus* OF44 have primarily focused on its genomic features and probiotic potential, whereas its effects on allergic diseases and host immune regulation remain largely unexplored.

In this study, we evaluated the therapeutic potential of *L. rhamnosus* OF44 in an ovalbumin-induced AR rat model. Specifically, we investigated whether oral administration of *L. rhamnosus* OF44 could alleviate allergic symptoms and restore immune homeostasis by modulating key T-cell subsets and cytokine responses involved in AR pathogenesis. Furthermore, metagenomic sequencing of the gut microbiota was performed to explore microbial compositional and functional alterations associated with the probiotic intervention. By integrating immunological profiling with microbiome functional analysis, this study aims to elucidate the role of *L. rhamnosus* OF44 in modulating the gut microbiota–immune axis and to provide mechanistic evidence supporting its potential application as a probiotic intervention for AR. To our knowledge, this study provides the first integrated evaluation of the immunomodulatory and microbiome functional effects of *L. rhamnosus* OF44 in AR.

Results

L. rhamnosus OF44 alleviated allergic symptoms in AR rats

Repeated OVA sensitization and challenge successfully induced AR in rats, as evidenced by significantly elevated behavioral scores for sneezing, nasal rubbing, and nasal secretion compared with controls (Figs. 1B and S1D–F). Treatment with *L. rhamnosus* OF44 at low, medium, or high doses markedly reduced these allergic symptoms, with effects comparable to those of loratadine. There were no significant differences in body weight between experimental groups throughout the entire trial period, nor were there any significant changes in the liver weight coefficient (Fig. S1A–B). The thymus and spleen indices were significantly increased in AR rats, reflecting systemic immune activation, whereas supplementation with *L. rhamnosus* OF44 significantly normalized both organ coefficients (Fig. 1C, D).

Histological analysis further confirmed the protective effects of probiotic treatment. H&E and PAS staining revealed marked epithelial injury, inflammatory cell infiltration, vascular congestion/edema, and goblet cell hyperplasia in the nasal mucosa of AR rats, all of which were substantially alleviated following administration of *L. rhamnosus* OF44 (Fig. 1E). Semi-quantitative histopathological scoring further demonstrated significantly elevated inflammatory scores in the AR group, whereas probiotic treatment markedly reduced the severity of nasal mucosal inflammation (Fig. 1F). Toluidine blue staining showed extensive mast cell accumulation and degranulation in the AR group, which were substantially reduced by probiotic intervention. Consistently, quantitative analysis confirmed a significant decrease in mast cell counts following *L. rhamnosus OF44* administration (Fig. 1G).

In peripheral blood, OVA induction increased leukocyte and neutrophil counts, while no significant change in eosinophil count was observed (Fig. 1H, I, Figure S1C). This finding suggests that OVA-induced AR in this model may primarily manifest as localized nasal inflammation rather than pronounced systemic eosinophilia. In addition, serum levels of IgA and IgG2a were decreased, whereas IgE, PAF-1, LTC-4, and IgG1 were elevated (Fig. 1J–O). These changes indicate a Th2-skewed humoral immune response. Treatment with *L. rhamnosus* OF44 significantly reversed these alterations, restoring IgA and IgG2a levels while suppressing IgE, PAF-1, LTC-4, and IgG1, indicating partial restoration of humoral immune homeostasis. To further characterize local immune responses at the site of allergic inflammation, we additionally quantified immunological markers in nasal mucosal tissues. Consistent with systemic findings, mucosal IgA levels were significantly decreased in the AR model group, whereas Th2-associated cytokines (IL-4, IL-5, and IL-13) and eosinophil cationic protein (ECP, eosinophil activation maker) were markedly elevated, indicating enhanced local type 2 inflammation and eosinophil activation. Notably, *L. rhamnosus* OF44 treatment significantly reversed these alterations (Fig. 1P–T). These results demonstrate that OF44 not only modulates systemic immunity but also effectively attenuates local nasal mucosal inflammation. Together, these results demonstrate that *L. rhamnosus* OF44 effectively mitigates allergic symptoms, reduces mucosal inflammation, and partially restores systemic immune homeostasis in OVA-induced AR rats.

L. rhamnosus OF44 modulated T cell differentiation and cytokine production in AR rats

The gating strategy for flow cytometry analysis is illustrated in Fig. 2A. Compared with the control group, AR rats exhibited a marked decrease in Th1 (IFN-γ^+CD4$^+$) cells and regulatory T (CD25$^+$Foxp3$^+$) cells (Fig. 2B, C), accompanied by a significant increase in Th2 (IL-4$^+$CD4$^+$) and Th17 (IL-17$^+$CD4$^+$) cell populations (Fig. 2D, E). Treatment with *L. rhamnosus* OF44 significantly restored the Th1/Th2 balance, reducing the proportion of Th2 and Th17 cells while increasing the frequencies of Th1 and Treg cells.

Consistent with these cellular findings, cytokine profiling demonstrated a similar trend. Serum levels of Th1- and Treg-associated cytokines (IFN-γ, IL-2, IL-12, TGF-β, and IL-10) were markedly reduced in AR rats (Fig. 2F–J), whereas Th2-associated cytokines (IL-4, IL-5, and IL-13) and pro-inflammatory mediators (IL-17, TNF-α, and IL-6) were significantly elevated (Fig. 2K–Q). Administration of *L. rhamnosus* OF44 significantly

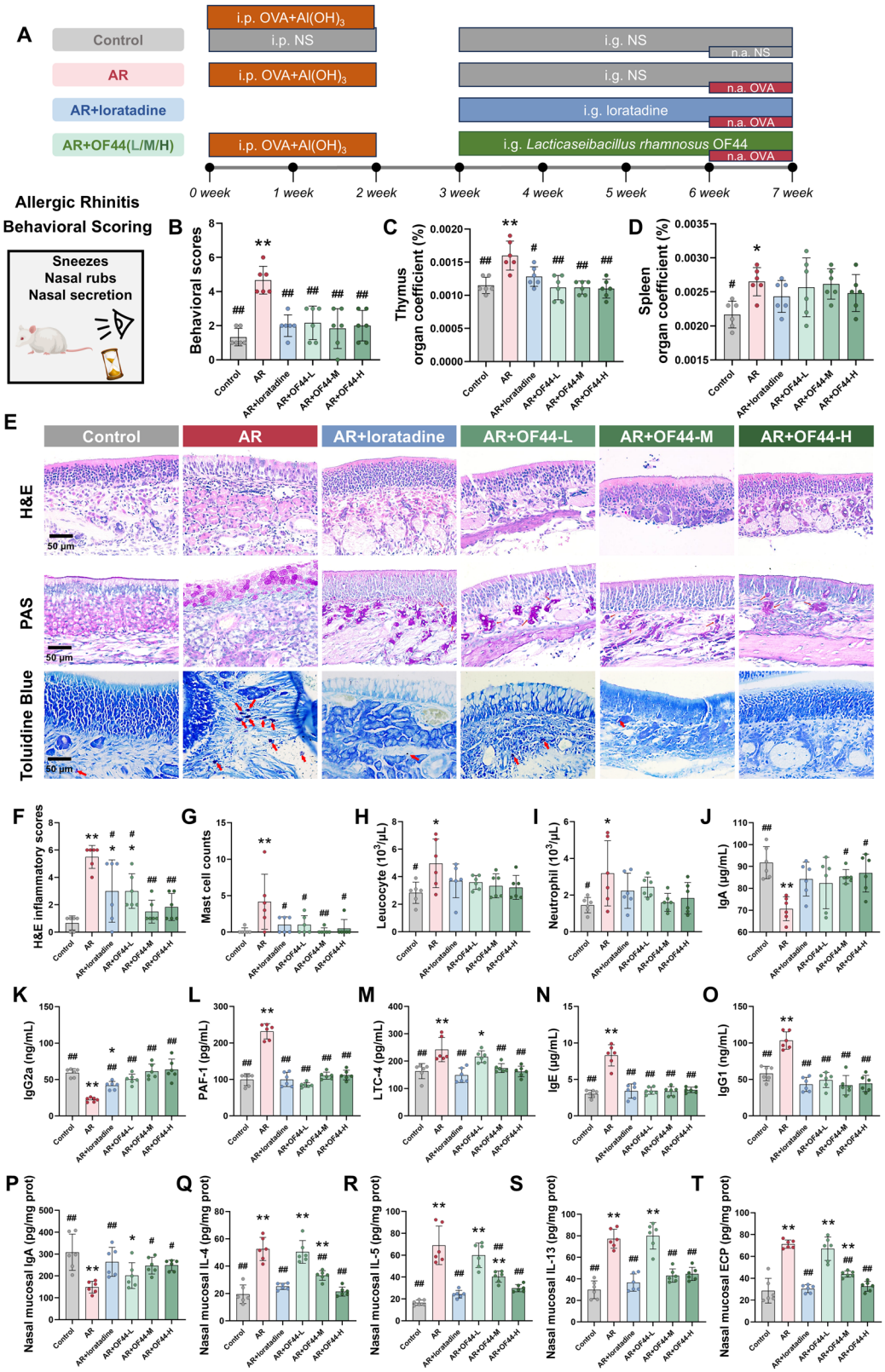


Fig. 1 | OF44 alleviates allergic symptoms, nasal mucosal inflammation, and allergic inflammatory mediators in OVA-induced allergic rhinitis. **A** Experimental grouping and procedures. **B** Rhinitis symptom scores. **C**, **D** Thymus and spleen indices. **E** Representative H&E, PAS, and toluidine blue staining of nasal mucosa. The red arrow indicates a mast cell. **F** Allergic inflammation scores based on H&E and PAS staining. **G** Mast cell counts based on Toluidine staining. **H**, **I** Blood leukocyte and neutrophil counts. **J**–**O** Serum levels of allergic rhinitis–related immunoglobulins and inflammatory mediators, including IgA, IgG2a, PAF-1, LTC-4, IgE, and IgG1. **P**–**T** Levels of nasal mucosal immunoglobulin, type 2 cytokines, and inflammatory mediators, including IgA, IL-4, IL-5, IL-13, and ECP. The data are presented as the mean ± SD, $n = 6$. One-way ANOVA followed by Tukey's post hoc test was used to analyze the differences between different groups. Compared to Control group: $^{*}P < 0.05$, $^{**}P < 0.01$; Compared to AR group: $^{\#}P < 0.05$, $^{\#\#}P < 0.01$.

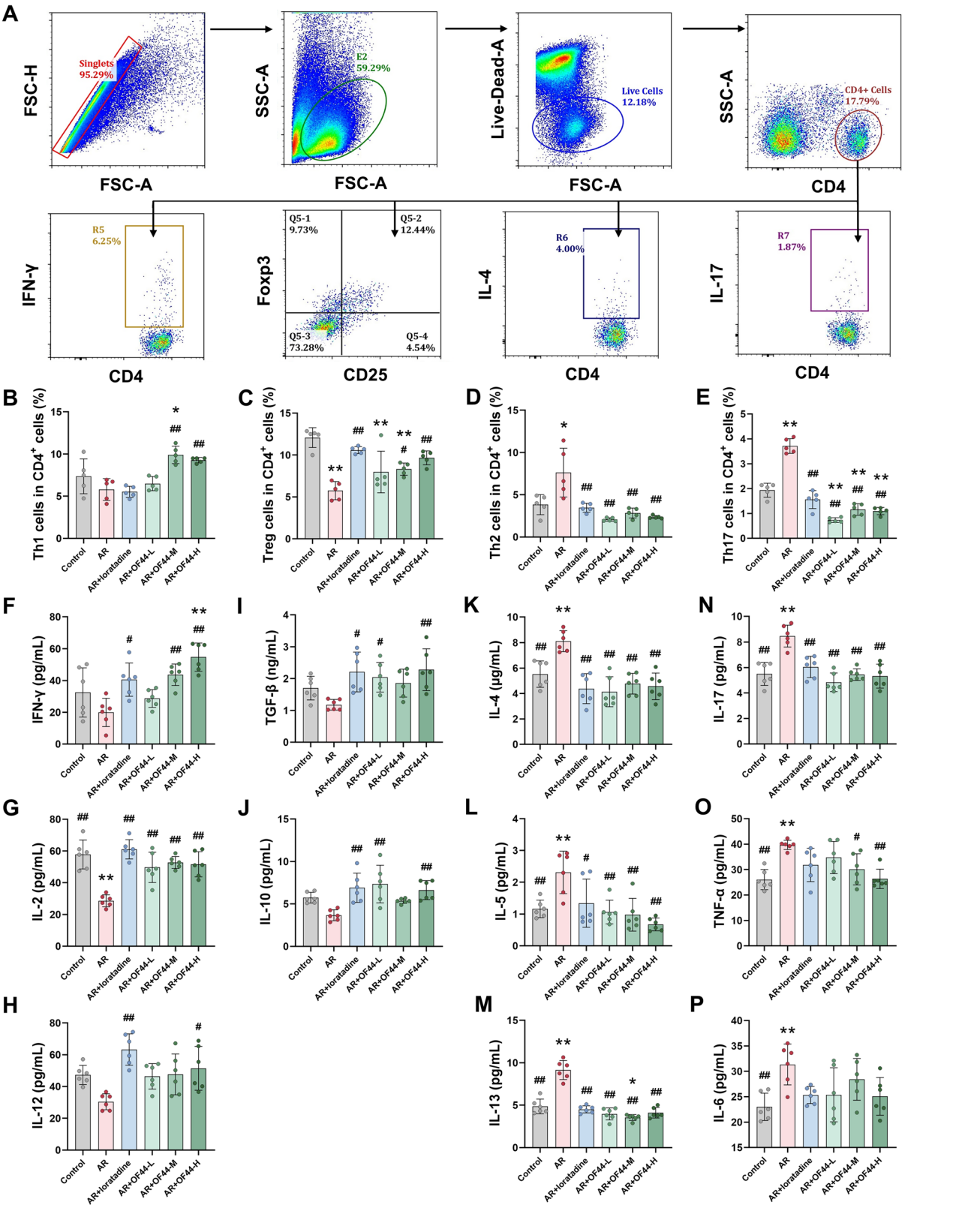


Fig. 2 | OF44 restores CD4⁺ T-cell subset balance and systemic cytokine profiles in OVA-induced allergic rhinitis. A Flow cytometry workflow. **B**–**E** Frequency of Th1, Treg, Th2, and Th17 cells within splenic CD4⁺ T cells. **F**–**H** Serum levels of Th1-related cytokines (IFN-γ, IL-2, IL-12). **I**, **J** Serum levels of Treg-related cytokines (TGF-β, IL-10). **K**–**M** Serum levels of Th2-related cytokines (IL-4, IL-5, IL-13). **N**–**P** Serum levels of Th17-related cytokines (IL-17, TNF-α, IL-6). The data are presented as the mean ± SD. The sample size for flow cytometry analysis (**B**–**E**) was $n = 5$ per group, and the sample size for serum cytokine measurements (**F**–**P**) was $n = 6$ per group. One-way ANOVA followed by Tukey's post hoc test was used to analyze the differences between different groups. Compared to Control group: $^{*}P < 0.05$, $^{**}P < 0.01$; Compared to AR group: $^{\#}P < 0.05$, $^{\#\#}P < 0.01$.

reversed these alterations, increasing the levels of IFN-γ, IL-2, IL-12, TGF-β, and IL-10, while reducing the production of IL-4, IL-5, IL-13, IL-17, TNF-α, and IL-6. Collectively, these results suggest that supplementation with *L. rhamnosus* OF44 alleviates allergic inflammation by restoring Th1/Th2/Th17/Treg immune homeostasis and rebalancing systemic cytokine responses.

L. rhamnosus OF44 restored gut microbial diversity and composition in AR rats

We first examined whether the intestinal barrier integrity was compromised in AR rats. Although H&E staining of rat colons showed no evidence that OVA sensitization disrupted the epithelial barrier (Figure S2A), serum levels of D-lactic acid were significantly increased. The lipopolysaccharide (LPS) content also showed an upward trend, although the difference did not reach statistical significance (Figure S2B–C). These findings suggest that OVA-induced AR may lead to subtle functional alterations in intestinal permeability without causing overt histological damage to the intestinal epithelium. Such changes are often referred to as functional or early-stage barrier impairment, which may not yet be detectable by conventional histological observation. However, additional analyses of intestinal barrier function, such as tight junction protein evaluation or permeability assays, are still needed for further validation. To explore the impact of *L. rhamnosus* OF44 supplementation on gut microbial ecology, metagenomic sequencing was performed to assess bacterial composition and diversity across groups. The species accumulation boxplot indicated that sequencing depth was adequate and sample size was sufficient (Figure S3A). At the phylum level, Bacillota, Actinomycetota, Bacteroidota and Pseudomonadota were dominant across all samples. Treatment with *L. rhamnosus* OF44, particularly in the medium- and high-dose groups (AR + OF44-M and AR + OF44-H), decreased the Bacillota/Bacteroidota ratio. This shift suggests that the overall microbiota structure shifted toward a more balanced composition (Fig. 3A and Figure S3B).

Alpha-diversity analysis revealed that AR rats exhibited reduced microbial richness and evenness, as indicated by lower Pielou evenness, Shannon, and Simpson indices, suggesting gut dysbiosis (Fig. 3B). These reductions were partially restored following probiotic supplementation. Beta-diversity analysis using PCoA and DCA demonstrated distinct clustering among groups, and PERMANOVA confirmed significant differences in microbial community structure ($R^2 = 0.2143$, $p = 0.01496$). The difference between the control group and the AR group was most pronounced, whereas *L. rhamnosus* OF44 intervention shifted the microbial composition toward that observed in the control group (Fig. 3C, D).

Linear discriminant analysis effect size (LEfSe) identified specific bacterial taxa enriched in different groups. The control group was enriched with taxa such as *Lachnospira*, *Blautia*, *Dysosmobacter*, and *Enterocloster*, many of which are beneficial commensals with anti-inflammatory properties, suggesting that these bacteria were depleted in the AR group. The AR group showed enrichment of *Escherichia* (classified within the order Enterobacterales and the family Enterobacteriaceae) (Figs. 3E–H and S3C–D). Members of this genus are often associated with inflammation and epithelial barrier dysfunction, and include opportunistic pathogens implicated in diseases such as urinary tract infections, gastroenteritis, and respiratory infections. Rats treated with the medium-dose probiotic exhibited an increased abundance of *Alistipes* (family Rikenellaceae) (Fig. 3F, G). Members of this genus are known to produce SCFAs and modulate the Th17/Treg cell balance, thereby influencing the immune system. Additionally, higher abundances of *Oscillibacter* and *Akkermansia* were observed in the low- and high-dose probiotic groups, both of which are associated with SCFA production and anti-inflammatory effects (Figure S4). Collectively, these findings indicate that *L. rhamnosus* OF44 supplementation partially reverses AR-associated gut dysbiosis by increasing microbial diversity and enriching beneficial commensal bacteria.

Functional shifts in gut microbiota are reversed by *L. rhamnosus* OF44 intervention

To further investigate the functional consequences of microbial compositional changes, we performed functional profiling based on the KEGG database. Ordination analyses, including PCoA and DCA based on KEGG orthologs (KOs), showed a distinct separation between AR and control groups, while administration of *L. rhamnosus* OF44 shifted the microbial function profiles toward those observed in healthy controls (Fig. 4A, B).

As the KEGG pathway level, the AR group showed enrichment of pathways related to biofilm formation, biosynthesis of unsaturated fatty acids, phenylalanine metabolism, and colicin antimicrobial peptide resistance, alongside suppression of pathways associated with amino acid metabolism, carbohydrate metabolism, and SCFA metabolism (Figs. 4C, D and S5). These alterations reflect a dysbiotic microbial functional state associated with pro-inflammatory activity. In contrast, treatment with *L. rhamnosus* OF44 markedly reversed these functional alterations, as evidenced by the downregulation of virulence-associated pathways (e.g., biofilm formation) (Figs. 4D and S5).

It is worth noting that metabolic pathways including histidine metabolism, arginine biosynthesis, tryptophan metabolism, and the citrate cycle (TCA cycle) were also restored in the AR + OF44-L, AR + OF44-M and AR + OF44-H groups (Figure S5). These changes indicate that *L. rhamnosus* OF44 treatment enhanced biosynthesis pathways associated with beneficial microbial metabolites, suggesting an improvement in the metabolic output of the gut microbial.

Compared with controls, AR rats exhibited enrichment of microbial gene modules associated with trans-cinnamate, pyrimidine and lysine degradation, purine biosynthesis, and multidrug resistance (Fig. 4E). Notably, several microbial gene modules enriched in the control group, such as those for histidine degradation, β-lactam resistance, and glycine cleavage system, were also enriched in the AR + OF44-M group, indicating partial functional restoration of the gut microbiota toward a healthy state (Fig. 4F). These findings indicate that supplementation with *L. rhamnosus* OF44 not only reshapes microbial community composition but also restores the functional metabolic capacity of the gut microbiota, thereby contributing to the attenuation of allergic inflammation.

Correlation analysis between microbial functional profiles and host immune/phenotypic parameters

To explore potential associations between probiotic-driven microbial alterations and host immune responses, we further examined the correlations between differential gut taxa, functional gene modules, and allergic phenotypes (Fig. 5 and Fig. 6). At the genus level, taxa such as *Deefgea*, *Amedibacterium*, and *Alkalicoccus* were positively correlated with nasal behavioral scores, serum IgE, IgG1, and Th2-associated cytokines (IL-4, IL-5, IL-13), suggesting that these taxa were associated with pro-inflammatory and allergy-promoting microbial signatures (Fig. 5A). In contrast, *Eleftheria*, *Luteimicrobium*, *Grimontia*, and *Pyxidicoccus* were positively associated with Treg-related cytokines (IL-10 and TGF-β) and negatively associated with IgE, LTC-4, PAF-1 (Fig. 5A). Network analysis further highlighted that IL-4, IL-5, IL-13, IgG1, IgE as key hubs nodes, suggesting that Th1/Th2 immune imbalance is strongly associated with microbiota variation in this model (Fig. 5B).

Functional correlation analysis revealed similar association patterns at the level of microbial gene functions. KEGG Ortholog (KO) modules associated with Th2 cytokines and inflammatory mediators included genes involved in LPS biosynthesis, biofilm formation, and secretion system pathways, suggesting microbial functional signatures commonly associated with inflammatory states. Conversely, several KO modules showed positive associations with IL-10 and TGF-β, and negative associations with IgE levels and nasal behavioral scores (Figs. 6A, B and S6A). These modules were mainly involved in amino acid metabolism (K05363, K18011), SCFA-associated pathways (K20626), and amino sugar and nucleotide metabolism (K01787, K03816). These patterns are consistent with microbial metabolic functions often linked to immune regulation and gut homeostasis. In

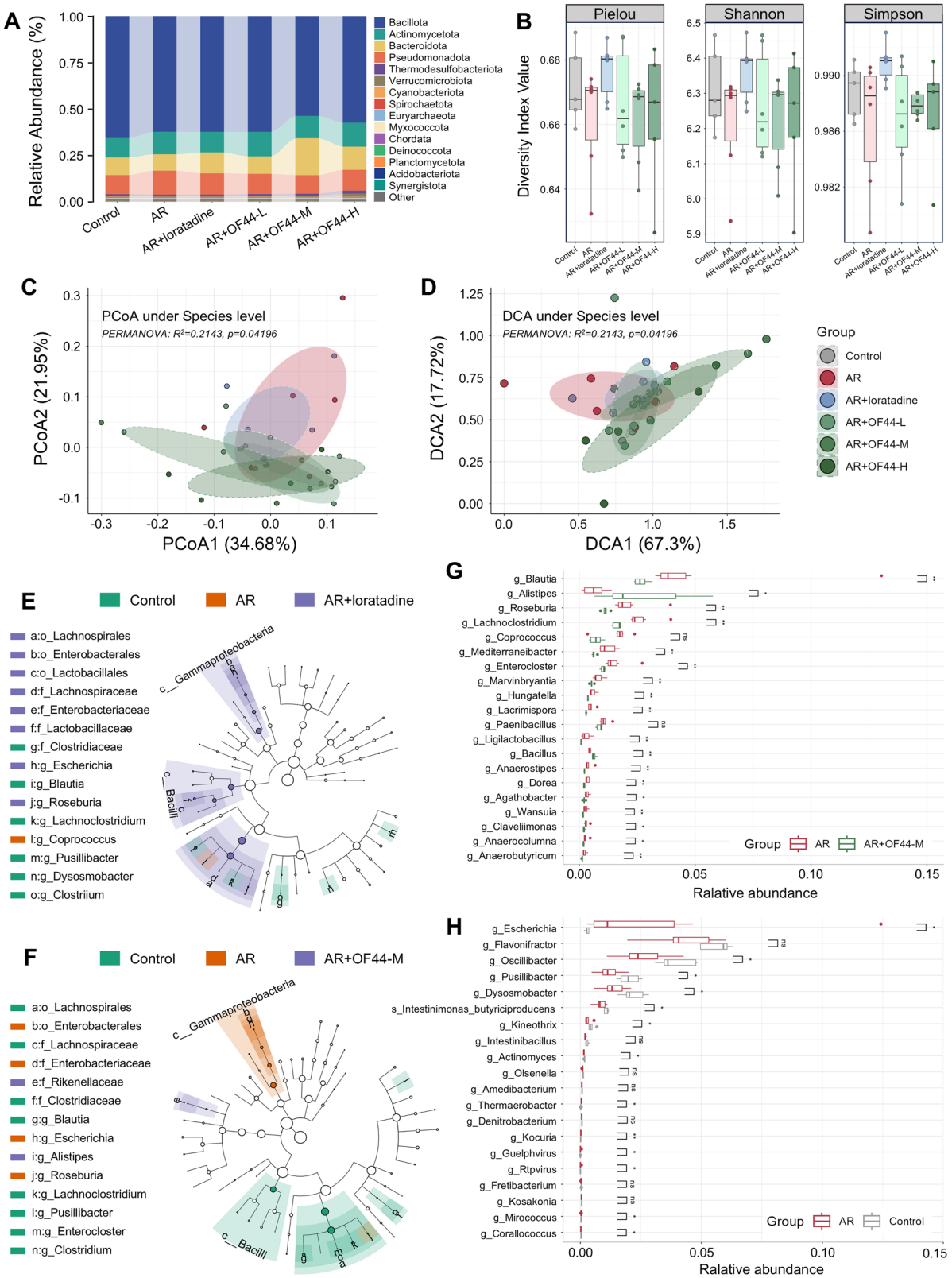


Fig. 3 | Metagenomic sequencing of gut microbiota in rats ($n = 6$). **A** Alluvial diagram at the phylum level. **B** α-diversity at the species level (Pielou, Shannon, Simpson indices). **C**, **D** β-diversity at the species level visualized by principal coordinate analysis (PCoA) and detrended correspondence analysis (DCA), based on Bray–Curtis distance matrices and assessed by PERMANOVA. **E**, **F** LEfSe evolutionary cladograms of differential species. **G**, **H** LEfSe box plots showing the top 20 genus with the highest relative abundance and statistical significance. Statistical significance between two groups in G–H was determined using Student's *t* test. $^{*}P < 0.05$, $^{**}P < 0.01$. Data are presented as mean ± SD.

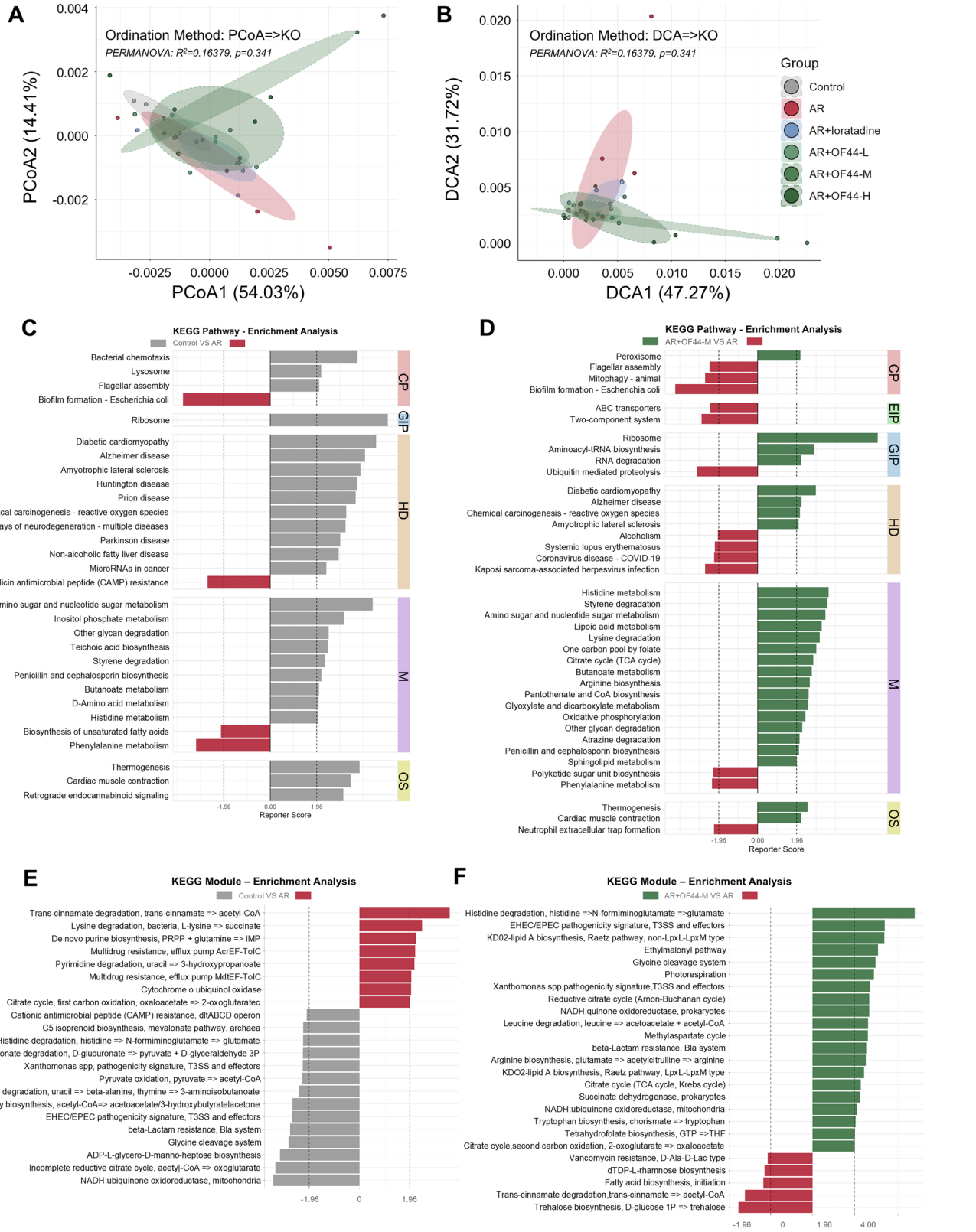


Fig. 4 | Functional annotation based on KEGG Orthology (KO) analysis to assess the effects of *Lacticaseibacillus rhamnosus* OF44 intervention on overall microbial functions and specific pathways ($n = 6$). A, B Bray–Curtis distance matrices derived from KO relative abundance tables visualized by PCoA and DCA, assessed by PERMANOVA. Differentially enriched functional pathways between groups were identified using the ReporterScore method. **C, D** ReporterScore results based on KEGG pathway. **E, F** ReporterScore results based on KEGG module.

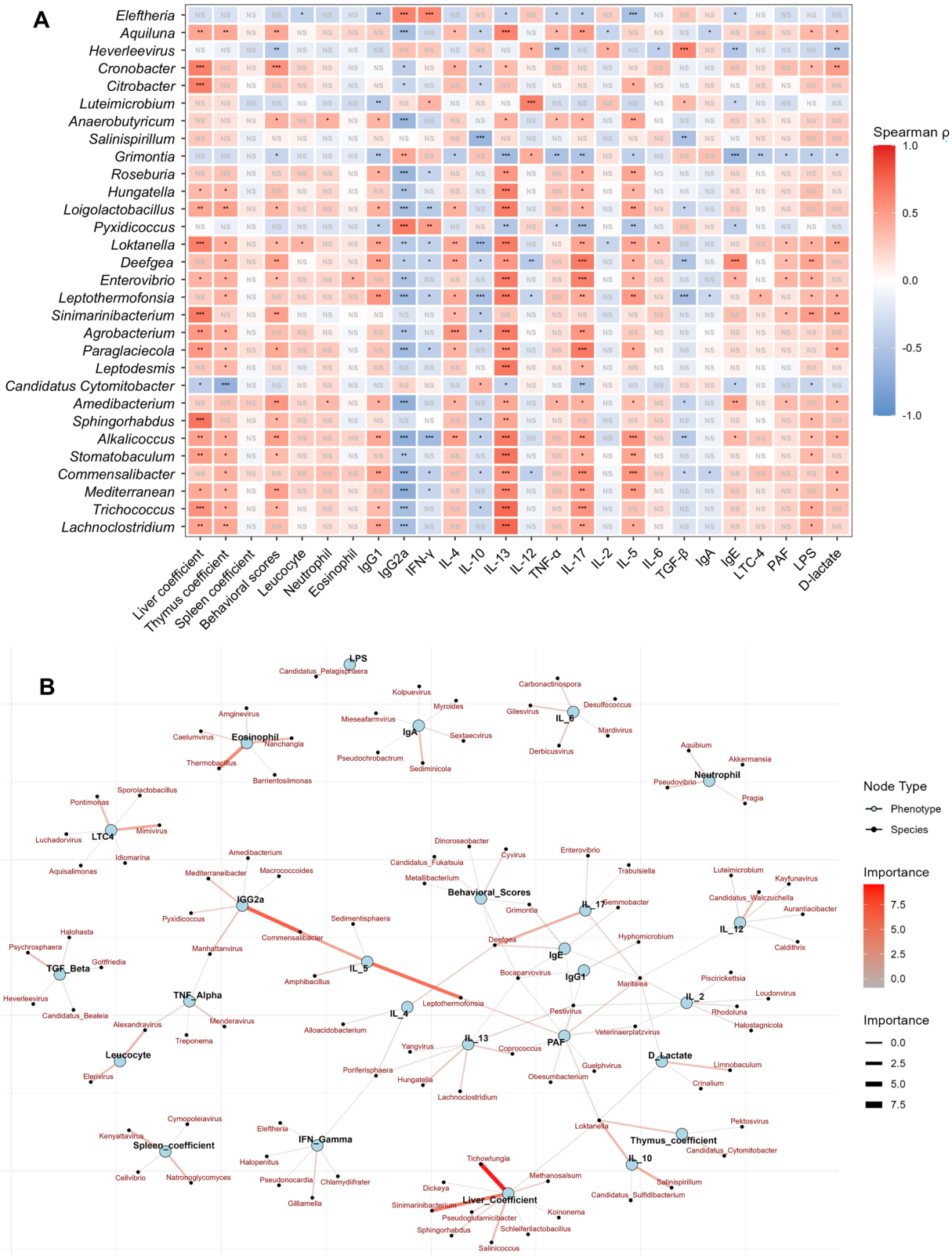


Fig. 5 | Association between microbial genus annotation and host phenotypes. **A** Heatmap of correlations between significantly altered genus abundances and phenotypic parameters based on Spearman's correlation analysis. The color scale represents Spearman's correlation coefficient (ρ). The significance of each correlation was assessed using a two-sided permutation test. Significance levels are indicated as $^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$. **B** Genus–phenotype association network constructed based on random forest importance analysis and visualized using the Fruchterman–Reingold force-directed layout algorithm. The network contains two types of nodes: filled circular nodes representing microbial taxa and hollow circular nodes representing host phenotypes. Edge width and color intensity were normalized to indicate the strength of the association (importance value). Data were derived from $n = 6$ biological replicates per group.

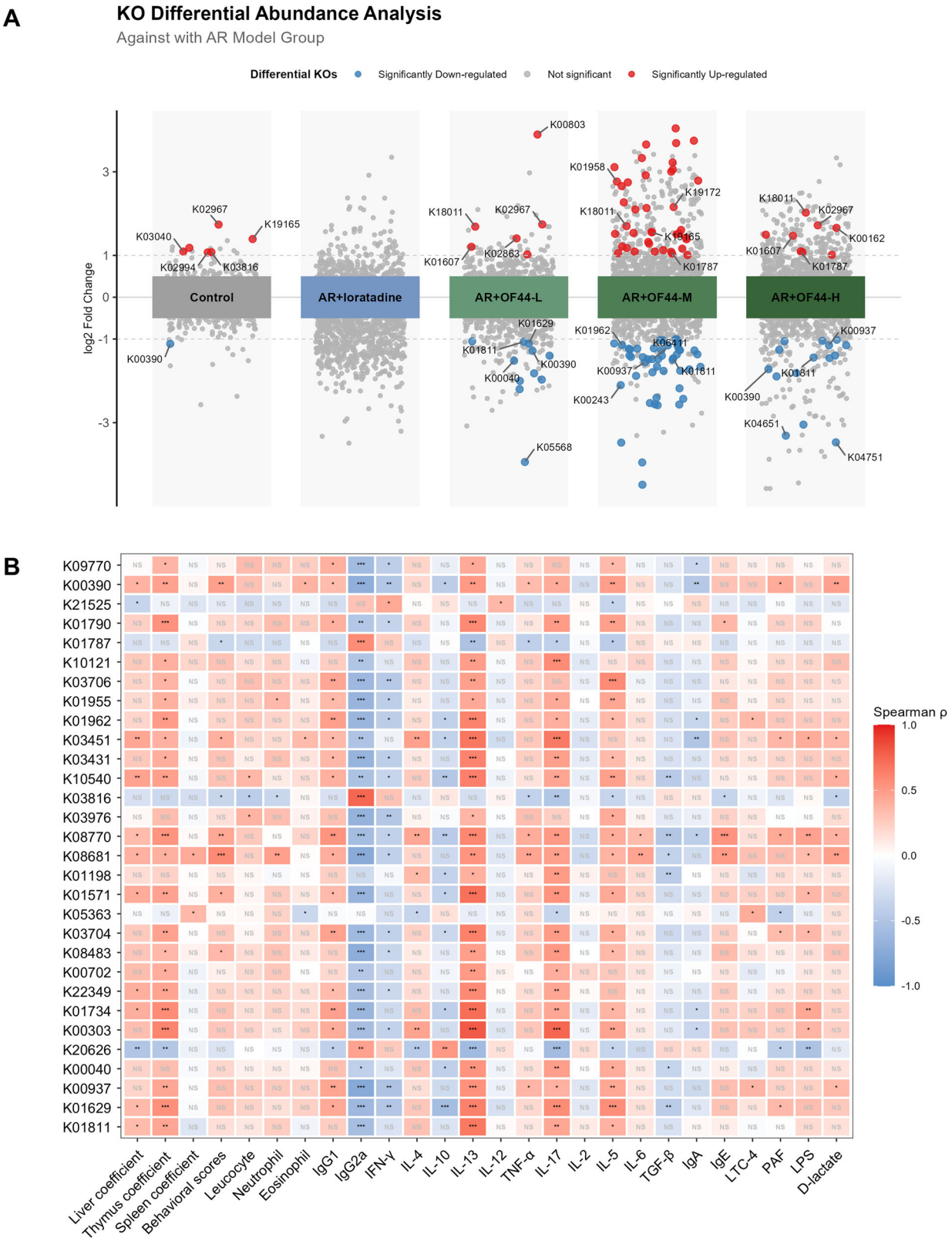


Fig. 6 | Association between functional annotations and host phenotypes. **A** Volcano plot of KO modules showing significant changes compared to the AR group. **B** Heatmap of correlations between significantly altered KOs (top 30) and phenotypic parameters based on Spearman's correlation analysis. The color scale represents Spearman's correlation coefficient (ρ). The significance of each correlation was assessed using a two-sided permutation test. Significance levels are indicated as $^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$. Data were derived from $n = 6$ biological replicates per group.

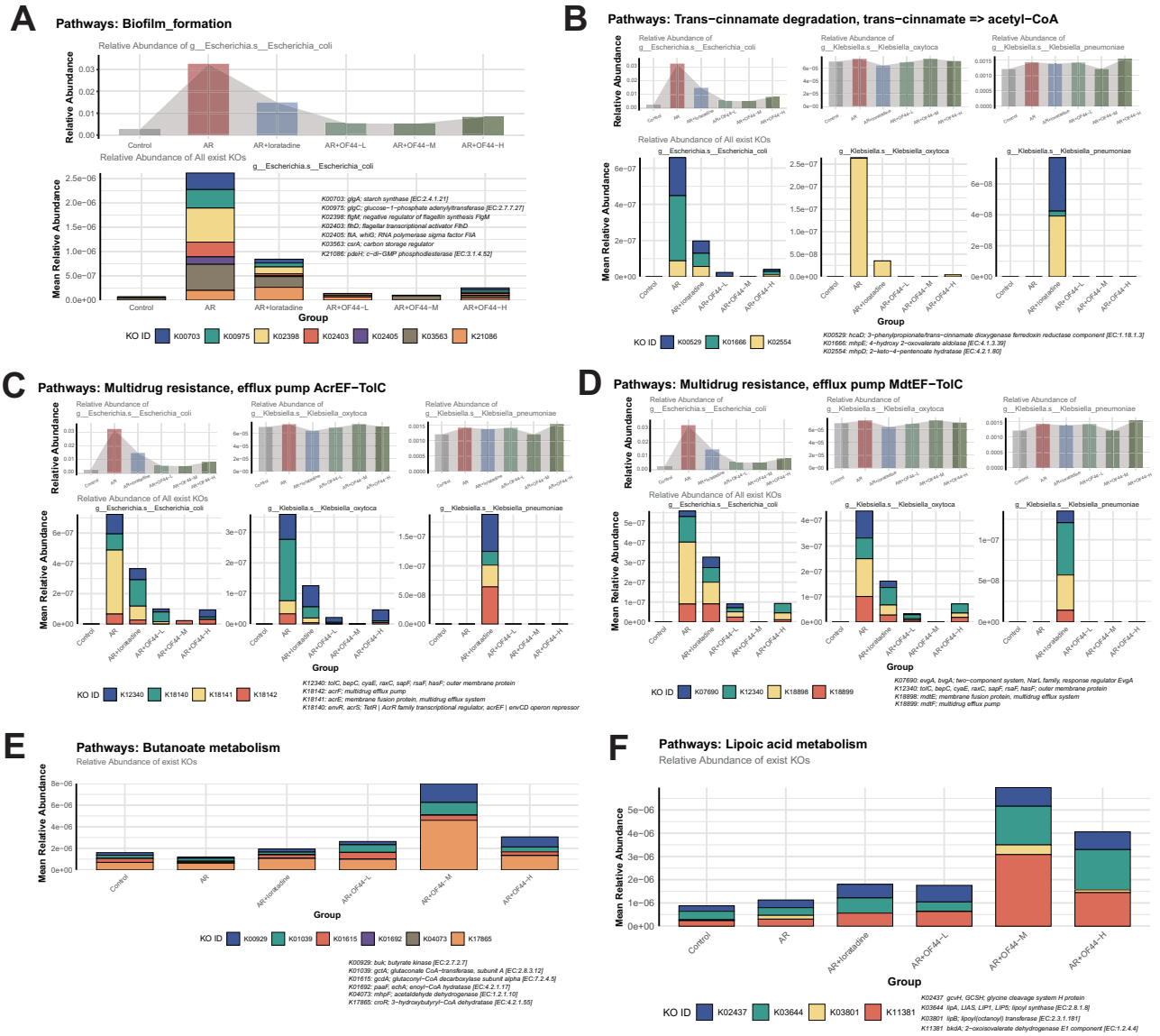


Fig. 7 | The microbial species abundance and KO profiling in AR-associated pathways. A Biofilm formation. **B** Trans-cinnamate degradation. **C**, **D** Multidrug resistance (two distinct modules). **E** Butanoate metabolism, and **F** Lipoic acid metabolism. **A**–**D** A two-layer stratification is shown: for each, the top bar plot displays the relative abundance of contributing microbial species, and the bottom stacked bar plot details their contribution to specific KOs. **E**, **F** present the functional decomposition (KO-level contributions) for pathways where species-level resolution was unclassified. Data were derived from $n = 6$ biological replicates per group.

addition, network analysis identified IL-4, IL-5, IL-13, IgG1, IgE, and PAF as major hub nodes, further supporting strong associations between microbial functional profiles and host immune parameters in the AR model (Fig. S6B).

Taxonomic and functional profiling of microbial contributions to AR-associated pathways and metabolic process

Building on previous findings, we further explored six AR-related pathways and metabolic processes, focusing on significantly altered KOs and their associated microbial taxa (Fig. 7 and Figs. S7 and S8). In the biofilm formation pathway, *Escherichia coli* enrichment in the AR group resulted in a significant increase in the relative abundance of key KOs (K21086, K03563) associated with biofilm formation and motility, potentially enhancing bacterial invasiveness. This was further supported by changes in flagellar biosynthesis-related KOs (K02405, K02403, K02398). In addition, the downregulation of K01666, K02554, and K00529 indicates a decreased capacity of trans-cinnamate degradation, implying suppression of microbial pathways linked to pro-inflammatory signaling (Fig. 7B). Fig. 7C, D demonstrate a marked reduction in the multidrug resistance pathway following *L. rhamnosus* OF44 treatment, indicating reduced antimicrobial resistance potential in opportunistic pathogens such as *Escherichia coli* and *Klebsiella oxytoca*. Moreover, key KOs involved in butyrate and lipoic acid metabolism were coordinately upregulated after *L. rhamnosus* OF44 intervention, suggesting enhanced microbial butyrate-producing capacity and improved metabolic stability of the gut microbiota.

Together, these integrated taxonomic and functional correlation patterns suggest that *L. rhamnosus* OF44 alleviates AR through microbiota-mediated mechanisms. This effect involves not only modulation of microbial community composition but also reprogramming of microbial functional networks toward an immunoregulatory and anti-inflammatory state, consistent with suppressed Th2 responses and enhanced immunological tolerance.

Discussion

AR is characterized by a Th2-dominant immune response accompanied by mucosal inflammation and systemic immune dysregulation. In this study, administration of *L. rhamnosus* OF44 markedly alleviated nasal allergic symptoms, restored the balance of serum and nasal immunoglobulin, and reestablished nasal mucosal integrity in OVA-induced AR rats. These findings suggest that *L. rhamnosus* OF44 may exerts multi-level immunomodulatory effects that extend beyond local nasal tissues and are associated with gut microbial remodeling and metabolic reprogramming.

Mechanistically, our data reveal that supplementation with *L. rhamnosus* OF44 corrected the Th1/Th2/Th17/Treg imbalance underlying AR pathogenesis. AR rats exhibited elevated Th2 (IL-4$^+$, IL-5$^+$, IL-13$^+$) and Th17 (IL-17$^+$) cell populations alongside decreased Th1 (IFN-γ$^+$) and Treg (CD25$^+$Foxp3$^+$) cells, consistent with previous observations in both patients and animal models of allergic airway inflammation[22,23]. Treatment with *L. rhamnosus* OF44 reversed this immune skewing, promoting Th1 and Treg differentiation while suppressing Th2 and Th17 expansion. These cellular changes were paralleled by restoration of nasal and systemic cytokine homeostasis, including increased levels of IFN-γ, IL-2, IL-12, TGF-β, and IL-10, and reduced levels of IL-4, IL-5, IL-13, IL-17, TNF-α, and ECP, indicating that the probiotic modulates both adaptive and regulatory immune circuits. A similar mechanism has been reported for *Lactiplantibacillus plantarum* NR16, a probiotic strain known to strongly induce Th1 responses. When co-cultured with immune cells, NR16 induces robust production of IFN-γ and IL-12. Consistently, oral administration of NR16 reduces airway hyperresponsiveness and leukocyte infiltration in mice[24]. Correspondingly, several biochemical indicators associated with AR pathology, including IgA, IgG2a, PAF-1, and IgE, were also significantly improved following administration of OF44.

Interestingly, although OVA sensitization did not cause histologically detectable intestinal barrier damage, metagenomic sequencing demonstrated significant microbial dysbiosis characterized by reduced α-diversity, elevated Bacillota/Bacteroidota ratio, and enrichment of Enterobacteriaceae. At a finer taxonomic resolution, this expansion was mainly driven by *Escherichia spp*., particularly *E. coli*. These bacteria are known to thrive under inflammatory conditions and can amplify mucosal immune activation through lipopolysaccharide-mediated pattern recognition receptor signaling. Such activation tends to promote Th2-skewed immune responses rather than directly inducing structural barrier damage[25]. In contrast, supplementation with *L. rhamnosus* OF44 significantly restored microbial diversity and selectively enriched taxa within the Rikenellaceae family, primarily represented by the genus *Alistipes*. Several *Alistipes* species have been reported to exert immunoregulatory effects through the production of SCFAs and other metabolites, which promote regulatory T-cell differentiation and enhance epithelial immune tolerance[26,27]. The enrichment of *Alistipes* observed in the *L. rhamnosus* OF44-treated groups coincided with increased levels of Treg-associated cytokines and reduced Th2-mediated inflammation, suggesting a potential link between probiotic-induced microbial compositional shifts and systemic immune reprogramming in AR.

Furthermore, these compositional improvements were accompanied by a coordinated restoration of microbial metabolic function. KEGG-based functional profiling revealed that *L. rhamnosus* OF44 normalized multiple core metabolic pathways, including amino acid metabolism (histidine metabolism, arginine biosynthesis, and tryptophan metabolism) and central energy metabolism, particularly the tricarboxylic acid cycle. At the same time, OF44 suppressed pro-inflammatory and pathogenic functional modules associated with biofilm formation, flagellar assembly, and antimicrobial resistance. The restoration of histidine and tryptophan metabolic pathways may be particularly relevant to immune regulation. Microbially derived histamine and tryptophan catabolites (such as indole-3-lactic acid and kynurenine) have been shown to activate aryl hydrocarbon receptor and G-protein-coupled receptor 109 A signaling, thereby promoting IL-10$^+$ Treg expansion and suppressing Th2 cytokine production[28,29]. Similarly, recovery of SCFA-related pathways (e.g., butyrate and propionate biosynthesis) is consistent with the observed increase in TGF-β and IL-10, as SCFAs enhance Treg differentiation through histone deacetylase (HDAC) inhibition and metabolic reprogramming[30,31]. Notably, OF44 intervention also restored butanoate metabolism and lipoic acid metabolism, pathways closely linked to microbial energy homeostasis, redox balance, and host immune regulation. Together, these functional changes suggest that *L. rhamnosus* OF44 may reshape microbial composition and metabolic activity in a manner associated with improved immune homeostasis.

Correlation analyses between microbial functions and host immune parameters further supported this mechanistic link. In the AR model group, KOs associated with biofilm formation and flagellar assembly, predominantly contributed by *Escherichia. coli*, were significantly downregulated following *L. rhamnosus* OF44 intervention. Flagellar assembly and biofilm formation are well-established virulence determinants that enhance bacterial adhesion, persistence, and immune activation within the host, indicating that probiotic treatment reduced the invasive potential of pathogenic bacteria[32]. Trans-cinnamate has been shown to inhibit activation of the inflammation-related signaling pathway, thereby suppressing the expression of multiple pro-inflammatory cytokines, including TNF-α, IL-1β, and IL-6[33]. Consistent with this observation, the downregulation of trans-cinnamate degradation pathways in both the control and probiotic-treated groups suggests a potential attenuation of pro-inflammatory cytokine signaling, which may partially explain the biochemical improvements observed in this study. Furthermore, KOs associated with bacterial multidrug resistance were prominently enriched in the AR group, indicating that the gut microbiota in AR rats may exist in a stress-adapted or drug-resistant ecological state. Such a state could enhance bacterial colonization fitness and aggravate microbial community imbalance, both of which were markedly reversed following probiotic supplementation. Importantly, several KOs involved in butyrate and lipoic acid metabolism were significantly upregulated after probiotic intervention, suggesting improved microbial metabolic stability and enhanced production of beneficial metabolites. Butyrate contributes to epithelial barrier maintenance, mucosal immune homeostasis, and anti-inflammatory responses, while lipoic acid supports cellular redox balance and metabolic enzyme activity[29,34]. The upregulation of these metabolic modules may therefore reflect enhanced microbial metabolic capacity, which could strengthen antioxidant defenses and metabolic resilience, while indirectly regulating host immune responses through interconnected SCFA–related metabolic networks[35]. Overall, these coordinated metabolic shifts suggest that OF44-associated microbiota remodeling may influence host immune responses through metabolite-mediated microbiota–host interactions[36].

In the present study, three probiotic doses of *L. rhamnosus* OF44 were evaluated (10^7, 10^8, and 10^9 CFU). All three doses significantly alleviated allergic symptoms and partially restored immune homeostasis, indicating that OF44 exhibits robust anti-allergic activity across a relatively broad dosage range. Interestingly, in several microbiome-related analyses, the medium-dose group (OF44-M) showed the most pronounced restoration of microbial diversity and functional pathways. This observation suggests that the probiotic effects of OF44 may not follow a strictly linear dose–response relationship. Instead, an intermediate dose may provide a more favorable balance for microbial colonization and host–microbiota interactions. Similar non-linear dose–response patterns have been reported in probiotic interventions, where the relationship between dose and host response is not strictly linear. For example, population-based analyses have demonstrated a U-shaped association between probiotic intake and health outcomes, suggesting that excessive intake may attenuate or even reverse beneficial effects[37]. In addition, meta-analyses indicate that increasing probiotic dosage does not always yield enhanced biological responses, potentially due to saturation of microbial colonization capacity or host–microbiota interactions[38]. These findings collectively suggest that optimal probiotic efficacy may occur within a specific dosage range rather than at maximal doses. Future studies will be required to further define the optimal dose range of OF44 under different host physiological and pathological conditions.

In addition, one limitation of the present study is that only the gut microbiota was profiled, whereas the nasal mucosal microbiome was not examined. Given emerging evidence that nasal dysbiosis contributes to AR pathogenesis, future studies integrating both gut and nasal microbiome analyses will be necessary to better characterize the microbiota–immune interactions along the gut–nasal axis. Furthermore, although our correlation and functional analyses suggest potential links between microbial alterations and host immune responses, these findings remain associative. Future studies employing germ-free animal models, antibiotic-mediated microbiota depletion, or fecal microbiota transplantation will be required to determine whether the immunomodulatory effects of OF44 are directly mediated through microbiota-dependent mechanisms.

Methods

Preparation for probiotic strain

L. rhamnosus OF44 was isolated from the fecal samples obtained from a healthy female donor. Genomic analysis revealed the presence of multiple genes associated with antimicrobial activity[39]. Both in vitro and in vivo studies have demonstrated that this strain can survive in the gastrointestinal tract of rats and has the potential to modulate gut microbiota composition[39,40].

The lyophilized powder of *L. rhamnosus* OF44 was kindly provided by BGI Precision Nutrition (Shenzhen) Technology Co., Ltd. Each gram of powder contained 2×10^{11} colony-forming units (CFU). For administration, the powder was reconstituted in sterile physiological saline, adjusted to the appropriate concentration for each experimental group, and administered to rats by oral gavage.

Experimental animals and ethics statement

Eight-week-old specific pathogen-free (SPF) Sprague-Dawley (SD) rats weighing 200 ± 20 g were housed in an environment maintained at 23 ± 2 °C with a 12-h light/12-h dark cycle and a relative humidity of 50 ± 5%. Rats had free access to irradiated sterile chow and purified drinking water. Drinking water and bedding were replaced every 2 days to maintain hygienic conditions and ensure experimental consistency. All experimental protocols were authorized by the Jiangsu Provincial Science and Technology Department (SYXK (Su) 2019-0056) and approved by the Science and Technology Ethics Committee of Nanjing University (IACUC-D2402106). All animals were handled in accordance with the ARRIVE guidelines and the requirements of European Directive 2010/63/EU.

Treatment details

After one week of acclimatization, rats were randomly assigned to six groups ($n = 6$ per group): a control group, an AR model group, a positive control group, and three probiotic supplementation groups (low-, medium-, and high-dose), receiving 1×10^7, 1×10^8, and 1×10^9 CFU per rat, respectively.

The experimental procedure consisted of four sequential phases: sensitization, immunization, intervention, and challenge (Fig. 1A).

In the sensitization phase, all rats except those in the control group received intraperitoneal injections of 1 mL of sensitizing solution containing 0.3 mg Ovalbumin (OVA; A5253-250G, Merck, USA) and 30 mg aluminum hydroxide adjuvant (239186-500 G, Merck, USA) every other day for a total of seven injections. This was followed by a 1-week immunization phase.

During the intervention phase, rats in the probiotic supplementation groups received 1 mL of probiotic suspension at the designated concentration daily by oral gavage. The positive control group received loratadine solution (0.9 mg/kg), and the control and model groups received an equal volume of physiological saline.

In the challenge phase, 50 μL of 5% OVA solution was instilled into each nostril of rats in the model and treatment groups once daily for seven consecutive days using a micropipette. On day 50, all rats were sacrificed, and blood, nasal mucosa, liver, spleen, thymus, colon tissue, and cecal contents were collected for subsequent analyses. The liver, spleen, and thymus were carefully isolated and weighed. The organ indices were calculated by normalizing organ weight to the final body weight of each rat using the following formula: organ index (%) = organ weight (g)/body weight (g) × 100.

Rhinitis symptom score

After the last nasal sensitization, symptoms were recorded over 10 min, including sneezing, nose rubbing, and nasal discharge. Symptoms were scored according to the criteria presented in Table 1.

Slice preparation

Nasal mucosa samples were fixed in 4% paraformaldehyde for 24 h, paraffin-embedded, and sectioned at 5 μm thickness. Sections were processed for hematoxylin and eosin (H&E), periodic acid-Schiff (PAS), and toluidine blue staining. Brightfield images were acquired using a full slide scanning system (VS200, Olympus). Allergic inflammation was scored according to the criteria presented in Table 2.

Table 1 | Rat rhinitis symptom scoring criteria

Symptom	Normal (0 score)	Mild (1 score)	Moderate (2 score)	Severe (3 score)
Sneezing	None	≤3 times	4–10 times	≥11 times
Nose rubbing	None	Occasional scratching	Frequent scratching	Vigorous rubbing
Nasal discharge	None	Contained within nostrils	Visible outside nostrils	Spread onto the face

Table 2 | Allergic inflammation in the nasal mucosa scoring criteria

Parameter	Normal (0 score)	Mild (1 score)	Moderate (2 score)	Severe (3 score)
Epithelial injury/ cilia loss	Intact epithelium with no cilia loss	Mild focal epithelial injury or slight cilia loss	Moderate epithelial injury with partial cilia loss or epithelial discontinuity	Severe epithelial damage with marked cilia loss or epithelial disorganization
Vascular congestion/ edema	No vascular congestion or edema	Mild vascular congestion or edema	Moderate vascular congestion or interstitial edema	Severe vascular congestion with marked edema
Inflammatory cell infiltration	No obvious inflammatory cell infiltration	Mild focal inflammatory cell infiltration	Moderate or multifocal inflammatory cell infiltration	Severe or diffuse inflammatory cell infiltration
Glandular/ mucosal hyperplasia	No glandular or mucosal hyperplasia	Mild glandular or mucosal hyperplasia	Moderate glandular or mucosal hyperplasia	Severe glandular or mucosal hyperplasia

Hematological analysis

For hematological analysis, at least 100 μL of whole blood was collected from the abdominal aorta of each rat into EDTA-coated anticoagulant tubes. The blood was immediately mixed with the anticoagulant by gentle inversion to prevent clotting. Total leukocyte, neutrophil, and eosinophil counts were subsequently measured using an automated hematology analyzer.

Immunological analysis

Blood samples were collected from the abdominal aorta and centrifuged at 3000 rpm for 15 min at 4 °C to obtain serum. Nasal mucosal tissues were homogenized in cold normal saline, and the homogenates were then centrifuged to collect the supernatants for subsequent enzyme-linked immunosorbent assay (ELISA) analysis. Kits were purchased from Jiangsu Meimian Industrial Co., Ltd and used according to the manufacturer's instructions to quantify the levels of IgE (MM-0063R1), IgA (MM-0062R1), IgG1 (MM-0062R1), IgG2a (MM-72014R1), IL-4 (MM-0191R1), IL-2 (MM-0192R1), IL-5 (MM-0094R1), IL-6 (MM-0190R1), IL-12A (MM-2342R1), IL-13 (MM-0085R1), IL-10 (MM-0195R1), IL-17 (MM-0088R1), IFN-γ (MM-0198R1), TGF-β (MM-0181R1), TNF-α (MM-0180R1), LTC-4 (MM-92804901), PAF-1 (MM-92804501), LPS (MM-926001O1), and D-lactic acid (MM-927258O1). ECP (LA166158H) were purchased from Nanjing Lapuda Biotechnology Co., Ltd.

Flow cytometry analysis

The proportions of $CD4^+$ Th1, $CD4^+$ Th2, $CD4^+$ Th17, and $CD4^+$ Treg cells in splenic single-cell suspensions were assessed, and their frequencies within the $CD4^+$ T-cell population were calculated. Rat spleen tissues were minced, filtered through a 70-μm cell strainer, and subjected to red blood cell lysis using RBC Lysis Buffer (10×; BioLegend, 420301) to obtain single-cell suspensions. Cells were then resuspended in RPMI 1640 medium (ThermoGibco, 11875093) supplemented with 10% fetal bovine serum and adjusted to 1×10^7 cells/mL.

For Treg analysis, cells were stained with BD Horizon™ Fixable Viability Stain 780 (BD, 565388), followed by surface staining with PE anti-rat CD4 (clone OX35; Thermo Fisher, 12-0040-82) and Alexa Fluor® 488 anti-rat CD25 (clone OX-39; BioLegend, 202108). Cells were then fixed and permeabilized using the Foxp3/Transcription Factor Staining Buffer Set (Thermo Fisher, 00-5523-00), followed by intracellular staining with APC anti-Foxp3 (clone FJK-16s; Thermo Fisher, 17-5773-82). Corresponding isotype controls were included, including Alexa Fluor® 488 Mouse IgG1, κ Isotype Control (BioLegend, 400132) and APC Rat IgG2a κ Isotype Control (Thermo Fisher, 17-4321-81).

For Th1, Th2, and Th17 cell analysis, splenic cells were stimulated with Cell Activation Cocktail with Brefeldin A (BioLegend, 423303) for 6 h at 37 °C in a 5% CO_2 incubator. After stimulation, cells were stained with BD Horizon™ Fixable Viability Stain 780 and surface-stained with PerCP/Cyanine5.5 anti-rat CD4 (clone W3/25; BioLegend, 201520). Cells were then fixed with Fixation Buffer (BioLegend, 420801), permeabilized with Intracellular Staining Permeabilization Wash Buffer (10×; BioLegend, 421002), and stained intracellularly with FITC anti-rat IFN-γ (clone DB-1; BD, 559498), PE anti-rat IL-4 (clone OX-81; BioLegend, 511906), and PE-Cyanine7 anti-rat IL-17A (clone eBio17B7; Thermo Fisher, 25-7177-82). Corresponding isotype controls included FITC Mouse IgG1, κ Isotype Control (BD, 554679), PE Mouse IgG1, κ Isotype Control (BioLegend, 400111), and PE-Cyanine7 Rat IgG2a κ Isotype Control (Thermo Fisher, 25-4321-82).

Flow cytometry data were analyzed using FlowJo software. Lymphocytes were first gated based on FSC and SSC characteristics, followed by exclusion of debris, doublets, and dead cells. $CD4^+$ T cells were then gated, and subset frequencies were calculated within the $CD4^+$ T-cell population. Th1 cells were defined as $CD4^+IFN\text{-}\gamma^+$ cells, Th2 cells as $CD4^+IL\text{-}4^+$ cells, Th17 cells as $CD4^+IL\text{-}17A^+$ cells, and Treg cells as $CD4^+CD25^+Foxp3^+$ cells. The $CD4^+$Th1/$CD4^+$Th2 and $CD4^+$Th17/$CD4^+$Treg ratios were also calculated.

Metagenomic sequencing

Cecal contents samples from rats were collected and immediately stored at −80 °C prior to DNA extraction. For metagenomic sequencing, cecal contents from five rats randomly selected from each group ($n = 6$) were used for downstream microbial composition and functional analyses. Metagenomic DNA was isolated using the MagPure Stool DNA KF Kit B (MAGEN, China) according to the manufacturer's protocol. The extracted DNA samples were aliquoted into DNase-free tubes (Axygen, USA) and stored at −20 °C under sterile conditions to avoid cross-contamination.

For metagenomic analysis, the constructed DNA libraries (MGIEasy Universal DNA Library Prep Set) underwent high-throughput sequencing using the DNBSEQ-T10 platform (BGI, Shenzhen, China). Sequencing was conducted in paired-end mode with a read length of 100-bp for all samples. Raw sequencing reads were first processed for quality control and adapter trimming using fastp (v0.23.4), with the following parameters: --length_required 70, --adapter_sequence AAGTCGGAGGCCAAGCGGTCTTAGGAAGACAA, --adapter_sequence_r2 AAGTCGGATCGTAGCCATGTCGTTCTGTGAGCCAAGGAGTTG. Subsequently, host read depletion was performed using Bowtie2 (v2.4.4), where the quality-processed reads were aligned against the rat reference genome GRCr8 (GenBank NCBI accession: GCA_036323735.1). Reads that mapped to the reference genome were discarded, and the retained non-host reads were used for subsequent taxonomic and functional profiling. Taxonomic classification was performed using Kraken2 (v2.1.3) with the Standard reference database and Bracken (v3.0.1), while functional profiling was analyzed using the HUMAnN3 (v3.8) pipeline.

Statistical analysis

Data were analyzed using GraphPad Prism 9 software (GraphPad Software Inc., USA) and presented as mean ± standard deviation (SD). Statistical analysis was performed using one-way analysis of variance (ANOVA) followed by Tukey's multiple comparison test. $P < 0.05$ was considered statistically significant.

Statistical analysis of microbial taxonomic composition was conducted through an integrated bioinformatics pipeline implemented in R (v4.5.0). Annotated taxonomic data were first normalized and transformed to ensure comparability across samples. The adequacy of sequencing depth was verified by species accumulation curves approaching an asymptote.

Within-sample (alpha) diversity was evaluated using Shannon, Simpson, and Pielou indices, with statistical significance assessed by Kruskal–Wallis tests followed by pairwise Wilcoxon rank-sum tests with Benjamini-Hochberg correction. Beta diversity was quantified using Bray–Curtis dissimilarity and visualized using ordination methods, including principal coordinates analysis (PCoA) and detrended correspondence analysis (DCA).

The significance of between-group differences was examined using permutational multivariate analysis of variance (PERMANOVA) with permutation testing. To identify differentially abundant genera across groups, LEfSe was employed.

Furthermore, associations between microbial taxa and host phenotypes were explored using Spearman correlation analysis, redundancy analysis (RDA), and random forest regression, followed by construction of microbial–phenotype interaction networks to identify potential relationships. All statistical tests employed a significance threshold of 0.05, with multiple-testing correction where applicable.

For microbial functional profiles, data were generated using HUMAnN3 for comprehensive annotation of gene families against the Kyoto Encyclopedia of Genes and Genomes (KEGG) database. Functional beta diversity based on Bray–Curtis dissimilarity was assessed using PCoA and DCA, with statistical significance determined by PERMANOVA.

Differential functional features were identified using ReporterScore analysis, and an empirical threshold of $|\text{ReporterScore}| > 1.96$ was applied to ensure statistical robustness.

Additionally, associations between microbial functional features and phenotypes were evaluated using an analytical framework parallel to that

applied for taxonomic data, including Spearman correlation analysis, redundancy analysis, and random forest regression. All associated KEGG Orthologs (KOs) within specific pathways were extracted based on the KEGG database and filtered to retain those showing significant intergroup differences in relative abundance.

Finally, the differential KOs were taxonomically assigned using stratified abundance profiles generated by the HUMAnN3 pipeline.

Data availability

The raw metagenomic sequencing data generated in this study have been deposited in the China National GeneBank Sequence Archive (CNSA) under accession number CNP0009160 and are publicly available.

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Author contributions

Liehai Hu: conceptualization, methodology, investigation, formal analysis, writing-original draft, visualization. Bin Hou, Songci Yan, Wenjun Tai, and Yunhui Xia: investigation. Jiang Wu: resources, writing-review & editing. Dongmei Li: resources, supervising, funding acquisition, writing-review & editing, project administration. Bin Shi: funding acquisition, project administration.

Competing interests

The authors declare no competing interests.

Additional information

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